

Computational supplement

1 + 3 covariant CMB anisotropies — analytical reproducibility bundle v1.0.0

1 Scope

This bundle is archived at [doi:10.25375/uct.34069509](https://doi.org/10.25375/uct.34069509), active on publication of v1.0.0.

This supplement accompanies the v1.0.0 analytical release. The bundle is an independent reconstruction, from first 1 + 3 covariant principles, of the almost-Friedmann–Lemaître results of Gebbie, Dunsby & Ellis, *Annals of Physics* **282**, 321 (2000). It reproduces; it does not argue. No new effect is claimed.

2 The acceptance spine

Appendix F of that paper, pp. 379–380, Eqs. (F.1)–(F.4), links the covariant mode recursion to external treatments written in real coefficients. Reproducing that chain tests the harmonic normalisation without introducing a representation.¹

Three external treatments are used. Wilson & Silk is not reproduced here: a coefficient transcribed through Appendix F would be the same source as Appendix F rather than an independent fourth.

3 Frames, and what v1.5.0 then is

The solution is carried with a **generic** u^a . The explicit frames appear as choices made at the end, not assumed at the start: the Newtonian frame $\tilde{\sigma}_{ab} = 0$, which is the specialisation Annals II itself makes at p. 365 in order to remove the shear terms, and the energy frame $q_a = 0$. Carrying the generic threading and specialising late is the difference the 1+3 covariant approach buys; a gauge-fixed treatment fixes the frame before the hierarchy is solved and cannot exhibit it.

This is also what fixes the status of the next release. Because v1.0.0 is the general case, **v1.5.0's restricted case is a recovery and a check** — it must return v1.0.0's numbers, and that requirement is the regression test rather than a new result.

Two consequences are visible in the figures. The decomposition of the temperature source into primary, Doppler and integrated terms (§7.1.1, Eqs. (177)–(179)) is frame-dependent, while the total is not, so it is drawn in two frames. And footnote 29 on p. 365 asserts that the Newtonian choice also removes the high- ℓ truncation difficulty of Appendix E; that is a checkable claim, and the convergence figure checks it in both frames.

4 Conventions

The normative sheet is carried, generated rather than copied, at `provenance/conventions.md`. Equation numbers in this bundle are the published numbers.

¹The comparison applies the $(2\ell + 1)^{-1}$ division that the text preceding (F.4) specifies and the display omits; (F.4) is otherwise correct as printed. A misprint, recorded in `provenance/ANNALS-II-ANTECEDENT-AND-CORRECTIONS-v1.md`.

The harmonic normalisation is $Q_{A_\ell} = (-k_{\text{phys}})^{-\ell} D_{\langle A_\ell \rangle} Q$. Both candidate conventions are run end to end; only this one reproduces the real, opposite-sign form of the external hierarchies.²

5 Comparison, and its range

The C_ℓ comparison is Sachs–Wolfe only, so its tolerance is bounded by the acoustic modulation dropped in reaching it. The range is derived, not chosen: $2 \leq \ell \leq 20$ at 5%, generated by `scripts/derive_ell_range.py`.³

6 Reproduction

```
python scripts/run_all.py --strict
```

7 Figures

²Annals I prints $(-1)^\ell$ at the corresponding step where its own definition gives i^ℓ . The covariant multipole τ_{A_ℓ} is invariant under the choice, so the misprint never produced a wrong result; the mode coefficients are not invariant, and the two statements are distinct. Both are asserted as tests.

³That derivation uses the Bessel identity of §8.3, which requires $[(m/2)!]^2$ in its denominator. As printed it is exact where $\Gamma(m/2 + 1) = 1$, so the large-scale result (201) is unaffected; the horizon-scale normalisation (204) and the D_ℓ of (205) are. A misprint, corrected silently and recorded.